A Trial to Evaluate the Safety and Systems Biology Response of Ebolavirus Zaire Vaccine (ChAd3-EBO-Z) in Healthy Adults

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Collaborator: University of Oxford — Jenner Institute
Vaccine Developer: GlaxoSmithKline (GSK) Biologicals SA
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1. Introduction and Background

Ebola virus disease (EVD), caused by Ebolavirus Zaire (EBOV), is a severe acute viral hemorrhagic fever associated with mortality rates of 25–90% in prior outbreaks. The 2013–2016 West Africa EVD epidemic—the largest in history—resulted in over 28,600 confirmed cases and 11,325 deaths across Guinea, Sierra Leone, and Liberia, prompting an unprecedented global public health response and accelerated vaccine development efforts.

ChAd3-EBO-Z is a replication-deficient chimpanzee adenovirus serotype 3 (ChAd3) vector expressing the full-length EBOV Zaire glycoprotein (GP), developed jointly by GlaxoSmithKline Biologicals SA and the Jenner Institute, University of Oxford. Prior Phase 1 data demonstrated acceptable safety across dose levels. This study evaluates safety at the optimized dose and applies multi-omic systems biology to characterize the innate and adaptive immune response landscape.

2. Study Objectives

2.1 Primary Objective

Evaluate the safety and tolerability of a single IM dose of ChAd3-EBO-Z (1×10^{11} viral particles [vp]) in healthy adults aged 18–50 years.

2.2 Secondary Objectives

- Characterize whole-blood transcriptomic (RNA-seq) responses at Days 1, 3, 7, 14, 28, and 180.
- Evaluate humoral immunity: anti-EBOV GP IgG binding antibody (ELISA) and pseudovirus neutralization titers (PsVNA).
- Assess T-cell responses by IFN-γ ELISpot at Days 14, 28, 56, and 180.
- Conduct plasma proteomics (mass spectrometry) and metabolomics (LC-MS/MS) at Days 0, 3, 7, and 28.
- Correlate early innate systems biology signatures with Day 28/56 adaptive immune response magnitudes.

3. Study Design and Schedule of Events

Phase 1, open-label, single-arm, single-center study conducted at the NIH Clinical Center (Bethesda, MD). Twenty healthy adult volunteers receive a single IM injection of ChAd3-EBO-Z (1×10^{11} vp) on Day 0 and are followed for 6 months.

Visit	Day	Key Procedures
Screening	–28 to –1	Consent, medical history, labs, eligibility screening, anti-ChAd3 baseline titer
Vaccination	0	Vaccination (1×10^{11} vp IM), safety labs, Day 0 blood (RNA-seq, proteomics, metabolomics)
Follow-up 1	1	Safety assessment; blood (RNA-seq)
Follow-up 2	3	Safety; blood (RNA-seq, proteomics, metabolomics)
Follow-up 3	7	Safety; blood (RNA-seq, immunology labs)
Follow-up 4	14	Safety; blood (ELISpot, serology: anti-GP IgG, PsVNA)
Follow-up 5	28	Safety; blood (ELISpot, serology, RNA-seq, proteomics, metabolomics)
Follow-up 6	56	ELISpot, serology (anti-GP IgG, PsVNA)
End of Study	180	Final safety assessment; ELISpot, serology, RNA-seq

4. Eligibility Criteria

4.1 Inclusion Criteria

- Healthy adults aged 18–50 years at enrollment.
- BMI 18.0–35.0 kg/m^2.
- Negative HIV-1/2 antigen/antibody combination test.
- Baseline anti-ChAd3 neutralizing antibody titer ≤1:200.
- Willing to forgo other vaccinations during the study period.
- No prior Ebola vaccination or confirmed Ebola virus infection.
- Females of childbearing potential must use effective contraception and have a negative pregnancy test at enrollment.

4.2 Exclusion Criteria

- Immunocompromising condition or immunosuppressive therapy.
- Receipt of blood products within 120 days.
- Positive serology for HBsAg or hepatitis C antibody.
- Significant cardiac, pulmonary, hepatic, or renal disease.
- Active or recent malignancy (excluding adequately treated non-melanoma skin cancer).
- High anti-ChAd3 neutralizing antibody titer (>1:200) that could ablate vaccine immunogenicity.
- Pregnancy, breastfeeding, or intent to become pregnant within 6 months.
- Current use of any investigational agent.

5. Vaccine Description and Administration

ChAd3-EBO-Z (lot # provided by GSK) is a frozen liquid formulation at 2×10^{11} vp/mL. A single 0.5 mL IM injection (1×10^{11} vp) is administered into the deltoid of the non-dominant arm on Day 0. Participants are observed for ≥60 minutes post-vaccination. Vaccine preparation and administration follow site-specific SOPs and GMP guidelines.

6. Outcome Measures

6.1 Primary Safety Outcomes

- Solicited local (pain, redness, swelling) and systemic (fever, fatigue, headache, myalgia, nausea) reactogenicity for 7 days post-vaccination (diary card).
- Unsolicited adverse events for 28 days post-vaccination.
- SAEs and medically attended AEs through Day 180.
- Safety labs (CBC with differential, CMP) at Days 0, 7, 14, and 28.

6.2 Systems Biology and Immunogenicity Outcomes

- Whole-blood RNA-seq transcriptomic profiles at all study visits.
- Anti-EBOV GP IgG ELISA at Days 0, 14, 28, 56, 180.
- Pseudovirus neutralization assay (PsVNA) at Days 0, 28, 56, 180.
- IFN-γ ELISpot (EBOV GP peptide pools) at Days 0, 14, 28, 56, 180.
- Plasma proteomics (data-independent acquisition LC-MS/MS) at Days 0, 3, 7, 28.
- Plasma metabolomics (untargeted LC-MS/MS) at Days 0, 3, 7, 28.

7. Statistical and Systems Biology Analysis

7.1 Safety Analysis

Safety outcomes summarized descriptively. Adverse event frequencies reported as proportions with exact 95% binomial CIs. Laboratory values graded using DAIDS Toxicity Grading Table (Version 2.1).

7.2 Transcriptomic Analysis

Differential gene expression analysis using DESeq2, comparing each post-vaccination timepoint to Day 0 baseline in paired contrasts. Gene set enrichment analysis (GSEA) applied with MSigDB Hallmark and Blood Transcription Module (BTM) gene sets. Multi-omics factor analysis (MOFA) integrates transcriptomic, proteomic, and metabolomic data to identify shared and modality-specific variance components. Pearson correlation between Day 3 systems biology module scores and Day 28 ELISpot / Day 28 antibody titers tests predictive signatures.

8. Ethical Considerations and Regulatory Compliance

The protocol is approved by the NIH IRB (Protocol #18-I-0120). The trial is conducted under NIAID IND 17208. Written informed consent is obtained from all participants prior to any study procedure. The study complies with the Declaration of Helsinki (2013 revision), ICH E6(R2) GCP guidelines, and applicable US federal regulations (21 CFR Parts 50, 56, 312).